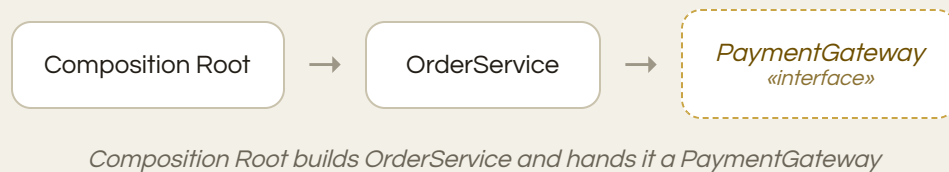


What Is Dependency Injection?

THE CORE IDEA, AND HOW IT RELATES TO INVERSION OF CONTROL



1. Depend on abstractions.

OrderService should hold a reference to a PaymentGateway interface, not a concrete StripeGateway.

2. Receive, don't construct.

Instead of `new StripeGateway()` inside OrderService, the concrete implementation is handed in from outside — typically as a constructor parameter.

3. Someone else decides.

A separate piece of code — a "composition root," or a DI framework/container — decides which concrete implementation to hand over.

DI VS. IOC — A COMMON MIX-UP

DI is one way to implement a bigger idea called **Inversion of Control**. Control over *which* implementation gets used is inverted — it moves out of the class itself and into whoever wires the app together.